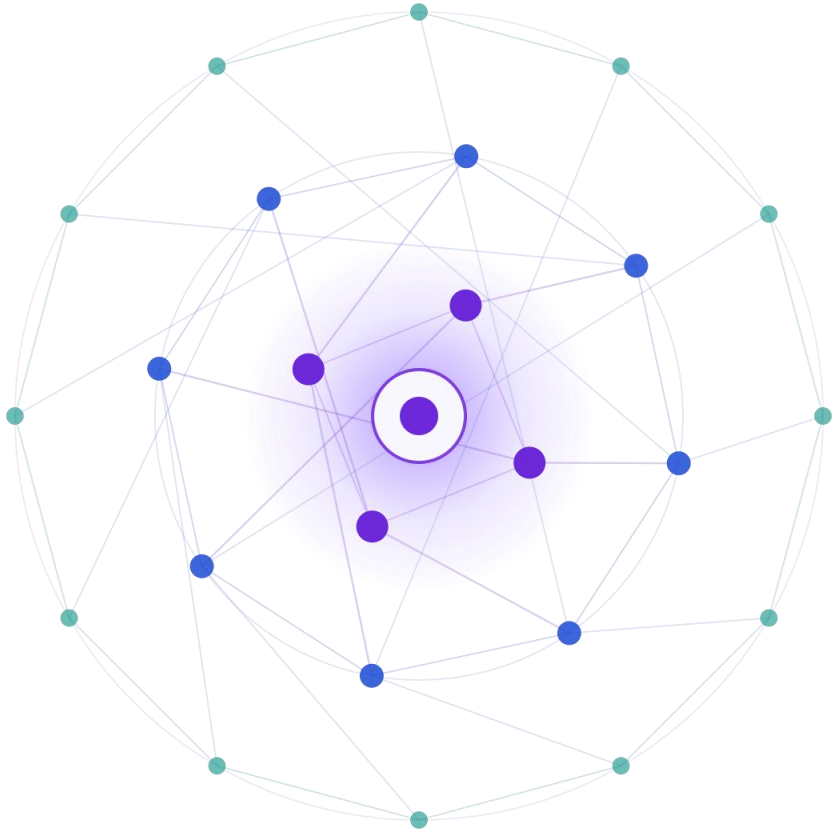
SAMSUNG PRISM

Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - 3
- Team Name – Cipheras
- College Name - SRMIST Ktr
- Member Name & Email 1- Aayushi & aayushimakker10@gmail.com
- Member Name & Email 2- Aditya Nair & adityanair5002@gmail.com
- Submission Github link - https://github.com/Adinair01/Teachable-Voice-Automation



Theme



Theme 3: Teachable Voice Automation



Most apps don't talk to assistants like Bixby, they need the app's own support and most apps don't add it .



Our Job : Build an assistant that learns any task by watching you do it once , then do it again when you ask by voice .



The gap is not a missing feature , it is a missing bridge between what a person says and what the app understands



We built that bridge as a layer on the top of the phone not inside any single app.

Existing Solutions & Gaps

- Bivby/Google Assistant : it only work with the apps that build in support for them
- Deep links : cover a handful of fixed actions , break the moment an app changes
- App SDKs : need the app's developer to add support , most won't
- Nobody lets you just show your phone a task once and have it repeat it, in any app without the app's help

Our Solutions & Architecture Diagram

Show it once , say it any way you like , it learns the task and repeats it in any Android app

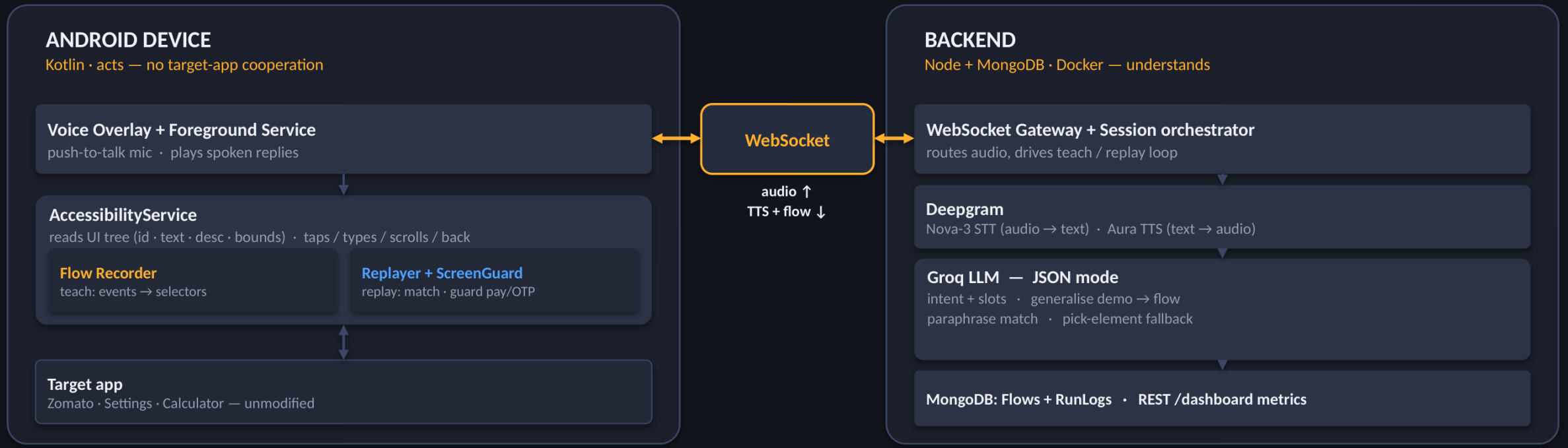
Built on Android's Accessibility Service , the same layer screen readers use to see and control a screen . No SDK , no app cooperation.

How it works : you do the task once -> we save it as steps (not screen coordinates , so it survives a different phone or a moved button) -> later

You ask by voice , in your own words -> it finds the task , fill in whatever changed and runs it -> stops right before payment or login and hands control back to you

Architecture

One WebSocket link: the phone acts through Accessibility, the backend understands with STT and an LLM.



How a command flows

Teach Replay

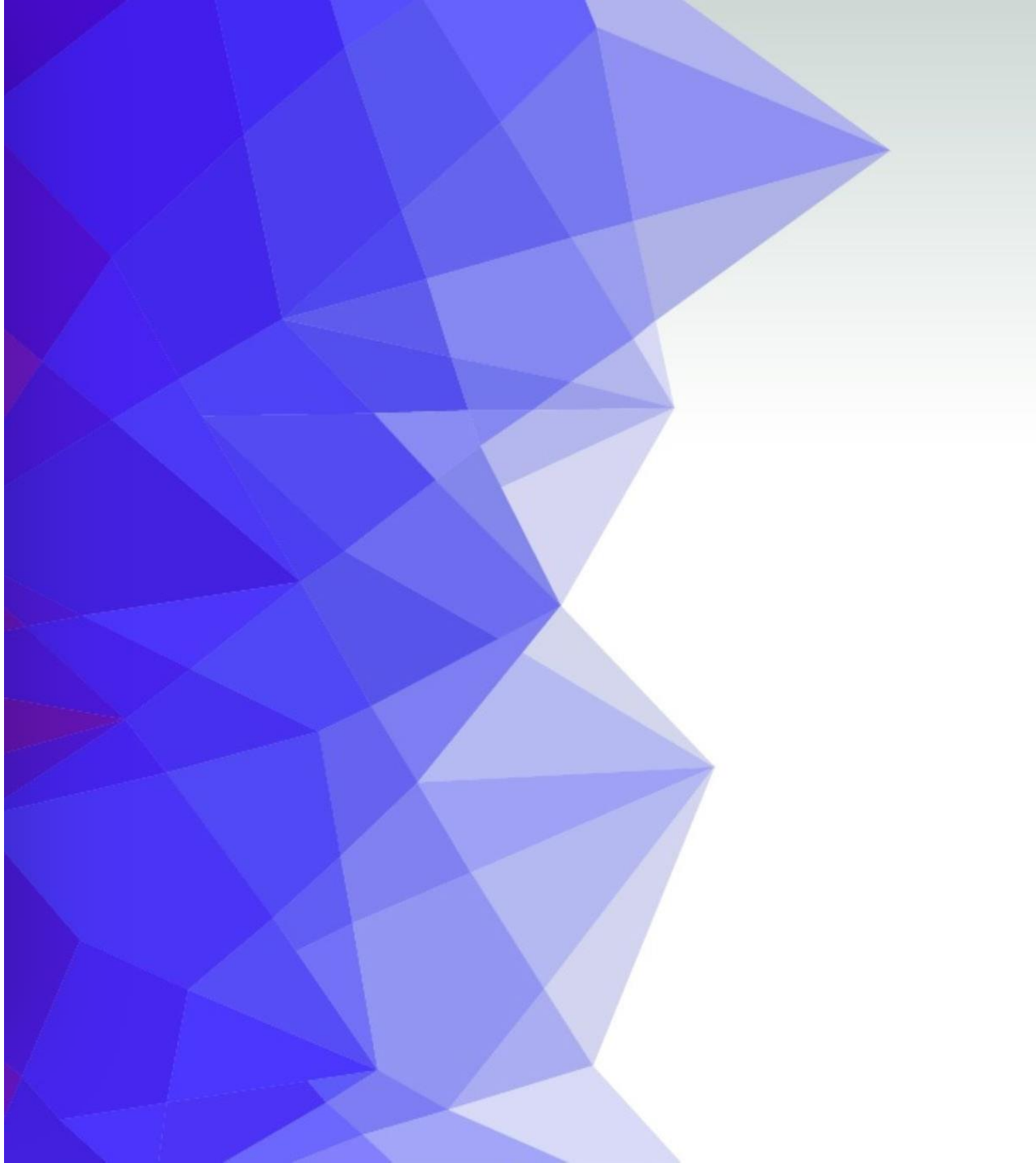
TEACH	Speak an unknown command	Do the task once by hand	Recorder → steps + selectors	Groq generalises → Flow with {{slots}}
REPLAY	Speak it with new values	STT · paraphrase match · slots	Replayer acts on live UI tree	ScreenGuard stops at pay/OTP → spoken result

Demo & Product Walkthrough

- Youtube Video Link - https://youtu.be/Mcz7sx1gQbA

Tools and tech stack used

- Android app: Kotlin, Accessibility Service
- Speech in: Deepgram (speech to text)
- Understanding: Groq / Llama, figures out what you meant, matches it to a learned task and fills in the details
- Speech out: Deepgram (text to speech)
- Backend: Node.js, Express, MongoDB, Docker



Impact & Use case

- Anyone who repeats the same phone task often such as ordering food , adjusting settings
- Especially useful for people who find typing or navigating apps hard like older people , users with limited vision or mobility
- Works with the apps exactly as they are today , no waiting on any developer
- Could grow into a system level assistant on the phone , not tied to one app

Innovation highlights, results and limitations

- Highlights : - learns by watching , not hardcoded
  - works across totally different apps without separate code for each
  - never touches a payment or login screen
- Results : -11 out of 14 required tests verified live on a real zomato account
  - Never once tapped past the payment screen , across every single run
  - Found and fixed 5 real bugs during live testing on a production app
- Limitations : it needs one clear demo to learn from ; ask a question when screen changes in a way it hasn't seen , not yet tested in every kind of app

What’s next

1. Teach it a task in a different kind of app to prove it generalises further
2. Have it ignore accidental interruptions while learning like a call , a notification
3. Handle more kinds of missing details mid flow , not just a few
4. Turn it into a background system service instead of an app you open

Brownie points

- No SDK, no Deep Link , no app permission , it works on any app today
- Remembers what to tap not where and also survives a different screen size or OTP
- A real safety boundary, hands back control at every payment login or OTP screen , no exceptions
- Tested live against a real production app not a toy dmo

Checklist- Updated on Public GitHub

- Working prototype code — public or shared GitHub repo (Y)
- README with reproducible setup instructions (Y)
- Demo video, max 5 minutes (YouTube video)
- Presentation file (PPT or PDF) (Y)

Thank you

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India